

Mutual Fund NAV Analysis (Programming Assessment)

Instructions

You are required to complete the following tasks by developing a **console-based application** in the programming language of your choice. All tasks must be performed using the mutual fund NAV data that you download as described below.

Task 1: Data Collection

- Download the last **7 years of regular growth options NAV data** for **any 5 mutual funds** from the **AMFI** website.
 - The data may be downloaded manually and must be saved locally in either **Excel (.xlsx)** or **CSV (.csv)** format.
 - The file must contain exactly **three columns: Fund Name, Date, NAV**.
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Task 2: Data Ingestion

- Develop a console application that **reads and processes** the NAV data from the Excel or CSV files.
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Task 3: Analysis and Computation

Implement separate functions/methods to perform the following operations:

1. Top-Performing Mutual Funds

- Calculate the **CAGR (Compound Annual Growth Rate)** for each mutual fund over the last **7 years**. You can find CAGR mathematical formula on internet
- Identify and display the **top 2 funds** with the **highest CAGR returns**.

2. Worst-Performing Mutual Funds

- Based on the same 7-year CAGR calculation, identify and display the **2 worst-performing funds** with the **lowest returns**.

3. NAV Swing Detection

- Identify all dates where a fund's **NAV changed by more than $\pm 5\%$** compared to the **previous available NAV value**.
- Display the **fund name, date, and percentage change** in the console.

Submission Requirements

- A **Git repository URL** containing the complete project. Must be public repo
- The project must **compile successfully** without errors.
- A **README.md** file that includes:
 - Step-by-step **instructions on how to run the program**.
 - Details of any **installations, dependencies, or setup steps** required.
- The **input data files** (Excel/CSV) used for analysis.